

LJA -- Requirements_status

Learning Journey Assistant — Requirements Status

Functional requirements

Non-functional requirements

Open questions from the August meeting, as of October

Learning Journey Assistant — Requirements Status

Version	1.0, 3 October 2026
Purpose	Status of every functional and non-functional requirement in LJA -- Requirements_for_SILO_analysis.pdf (the team's record of the 11 August 2026 meeting with the project owner) against main at b4ccebd. That PDF is a source document and is left as written; this page is the trace.
Reading the status	Met: on main with evidence. Partly: a bounded slice exists; the limit is stated. Not built: with the reason. Superseded: the requirement was replaced by a recorded decision.
Source	docs/design/requirements-status.md; rebuild with docs/design/build.sh

Functional requirements

ID	Requirement (abridged)	Status	Where, and what to know
FR-1.1	Parse the workbook's three tabs into a normalised model	Met	<code>lja.data.excel_loader</code> → <code>LjaDataset</code> ; a malformed workbook is an error
FR-1.2	Assessment to SILO is one-to-many	Met	<code>Assessment.silo_ids</code> ; one score counts towards every SILO it covers (documented approximation)
FR-1.3	Idempotent, re-runnable ingestion	Met	Pure in-memory load; no database writes
FR-1.4	Source adapter interface; spreadsheet and Moodle populate the same model	Met	<code>--source excel\ moodle</code> on every command; the Moodle loader was added with no change to clustering or gap detection
FR-1.5	Pseudonymise student id at ingest	Partly	Not at ingest: ids stay as supplied (synthetic <code>STU0001</code> -style). Pseudonymisation exists at export (<code>lja.export --anonymise</code> , keyed HMAC). Before real data touches a cloud provider, pseudonymise at ingest (SMD §3.8)
FR-1.6	Reject malformed rows and report	Met	Loader raises on unmapped criteria and malformed sheets rather than dropping rows; no separate quality report
FR-1.7	Ingest a course map when supplied	Partly	The slot exists: <code>LJA_SUBJECT_SEQUENCE</code> orders subjects and <code>trajectory.py</code> labels its source. No course map has been supplied, so the default is the three supplied subjects and other subjects fall back to the year digit
FR-2.1	Stable SILO identity plus verbatim text	Met	<code>Silo(subject_code, silo_local_id, text)</code> , keyed <code>SUBJ:SILOn</code>
FR-2.2	Subject level and provenance	Partly	Year level read from the subject code; the catalogue carries provenance (source: <code>handbook\ synthetic</code>) but the workbook path does not
FR-2.3	SILO text immutable, versioned	Not built	No versioning; a re-cluster on changed wording produces new cluster ids (a hash of

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			members), which is the practical safeguard
FR-2.4	30+ subjects without schema change	Met	The 100-subject, 5,000-student cohort runs through the unchanged pipeline (<code>data-fixtures/README-handbook-cohort.md</code>)
FR-3.1	Semantic similarity across SILO pairs via embeddings and/or LLM	Met, by a different route	Direct LLM clustering of the whole suite (ADR 0002), not pairwise scores; the embedding tagger (<code>competency_tagger</code>) exists for the catalogue and is the staged route for the pipeline
FR-3.2	No keyword matching in production	Met	None used
FR-3.3	Configurable similarity threshold recorded per map	Superseded	No pairwise threshold exists under ADR 0002; the clustering cache records the model and the gate records decisions
FR-3.4	Every edge stores score, rationale, model, prompt version, timestamp	Partly	Each cluster stores a rationale; the model and prompt version are not written into the cache (debt item 3, A-29)
FR-3.5	Academic can accept, reject or adjust; overrides persist	Met, CLI only	<code>lja.review</code> : pending / confirmed / rejected with a note, kept beside the cache, surviving re-clustering for unchanged clusters. No browser UI
FR-3.6	SILO graph exportable for network visualisation	Partly	The dashboard draws the competency-to-subject-to-outcome trace and a chord diagram; <code>clusters.csv</code> is the flat export. No node/edge file
FR-3.7	Incremental recomputation	Not built	A new subject re-clusters the suite; the gate's decisions survive
FR-4.1	Cluster linked SILOs into competencies	Met	<code>silo_clustering</code> , with coverage validation and retry
FR-4.2	Label and describe each competency	Met	Label and rationale per cluster, reviewed by staff (IOLG-124)
FR-4.3	A SILO may belong to more than one competency	Superseded	Deliberately exactly one, so attainment is not double-counted; the grounding check enforces it
FR-4.4	Each competency resolves to SILOs, subjects, levels,	Met	The competency page's trace: competency → subjects →

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	assessments		outcomes → assessments
FR-4.5	Labels editable and lockable	Partly	Rejecting with a note is the edit path; no relabel-in-place
FR-5.1	Per-student competency score aggregated through assessment → SILO → competency	Met	<code>compute_gaps()</code> , weighted by assessment weight
FR-5.2	Pluggable aggregation, default unweighted mean	Met, by a different default	Weighted mean; weights of 1.0 reproduce the unweighted case
FR-5.3	Intra-student deficiency relative to the student's own profile	Met	Median and MAD within the profile, with absolute guards (ADR 0001); thresholds unratified (A-01)
FR-5.4	Inter-student deficiency against the cohort	Partly	Cohort statistics and priority groups show where a student sits; no per-competency cohort-relative classification, by decision (SMD D8)
FR-5.5	Track scores across levels 1, 2, 3 to expose trajectory	Met	<code>trajectory.py</code> : declared subject sequence, trend with its basis, subjects ahead (IOLG-106)
FR-5.6	Every flagged gap drills down to its assessments, SILOs and scores	Met	The gap card's evidence table and basis line; the Provenance page
FR-5.7	Thresholds configurable	Met	Seven <code>LJA_GAP_*</code> variables, CLI flags, shown on the Provenance page
FR-5.8	Reproduce the workbook's at-risk band alongside computed gaps	Met	Performance band on the student list and page; the dashboard's own at-risk cohort is deliberately undefined (D11)
FR-6.1 to 6.5	Feedback comment analysis	Not built	The supplied feedback is 45 templated strings across 1,650 rows; the owner flagged bespoke feedback as a re-identification risk. The synthetic generators (FR-6.4) exist and flag synthetic rows
FR-7.1	Ranked interventions per gap	Met	Learning plan priorities in order
FR-7.2	Three intervention classes: subjects, resources, quizzes/study material	Partly	Study strategies and practice quizzes exist; external resources and subject suggestions do not, and subject advice is excluded by the guardrails (SMD D19)
FR-7.3	Recommendations contextualised to the forward plan and course map	Partly	The trajectory names where a gap is assessed next; the

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			strategy's <code>prepare_for</code> names it; no course map yet
FR-7.4	Every recommendation cites its gap and evidence	Met	Grounding by code: every name must exist in the student's own context
FR-7.5	Trigger decoupled from the end-of-semester batch	Met	Each generator is its own command; the dashboard's opt-in button (PR #55) runs it on demand
FR-8.1	Assess SILO clarity, specificity, measurability, isolation	Met	<code>silos_quality</code> : flagged, orphan, unassessed, vague wording; the <code>/silos</code> page and lists
FR-8.2	Report for coordinators with justification and rewording	Partly	The lists are the report; no suggested rewording
FR-9.1	Export competency map, gaps and intervention log, CSV and JSON	Met, CSV	<code>lja.export</code> : three CSVs plus a JSON manifest; <code>in_plan</code> records whether a plan targets the competency
FR-9.2	De-identified by default; re-identification audited	Partly	<code>--anonymise</code> is an option, not the default; keyed HMAC, salt kept out of git
FR-9.3	Snapshot timestamp and configuration in the export	Met	<code>manifest.json</code> : source, commit, cache, thresholds, schema version

Non-functional requirements

ID	Requirement (abridged)	Status	Where, and what to know
NFR-1	Scale to 30+ subjects; vector retrieval before LLM adjudication; cache embeddings	Partly	Single-call clustering fails at 52 SILOs on the local model; the embedding tagger caches embeddings and scales (1,436 SILOs in 64 s) but is not yet the pipeline's clusterer (ADR 0002 review trigger)
NFR-2	Pseudonymised ids; no identifying text in prompts or logs	Partly	Prompts carry STU0001-style ids and marker feedback; real data would need ingest-time pseudonymisation (FR-1.5)
NFR-3	Model, prompt version and thresholds persisted with every artefact; temperature 0	Partly	Thresholds in the export manifest and on the Provenance page; model in the generation logs, not in the cache (A-29); temperature 0.2 by measured choice
NFR-4	Provider abstraction with an offline stub	Met	LLMClient protocol, two providers; the committed reference run is the offline path for the pipeline and dashboard, and tests use fake clients
NFR-5	Token spend bounded and reported	Met	Every command prints calls, tokens and time; the ten-plan audit cost \$0.62 on Anthropic
NFR-6	No automated claim without traceable evidence	Met	Classification basis on every row; grounding on every artefact; the quiz states the one thing it cannot verify
NFR-7	Local-first, no institutional network dependency	Met	Dockerised Moodle, local models, localhost dashboard; the chart libraries come from a CDN (debt item 1)
NFR-8	The feedback-cleansing constraint recorded as a limitation	Met	SMD §3.9 item 9; this page

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#	Question	Status
1	The rule behind the at-risk band	Confirmed there is none; the dashboard defines no at-risk cohort and the thresholds remain unratiſed (A-01)
2	A sanitised extract of bespoke feedback	Not supplied; feedback analysis not built
3	Course maps for CS, IT and Cyber	Not supplied; LJA_SUBJECT_SEQUENCE is ready for them
4	Preferred or prohibited LLM provider	Open; both providers are built and the choice is configuration
5	Institutional competency framework	Open; labels are derived from the data and reviewed by staff
6	Weekly meeting slot	Settled in Sprint 3